similarly split till we get to leaves, each of which contains just one character.

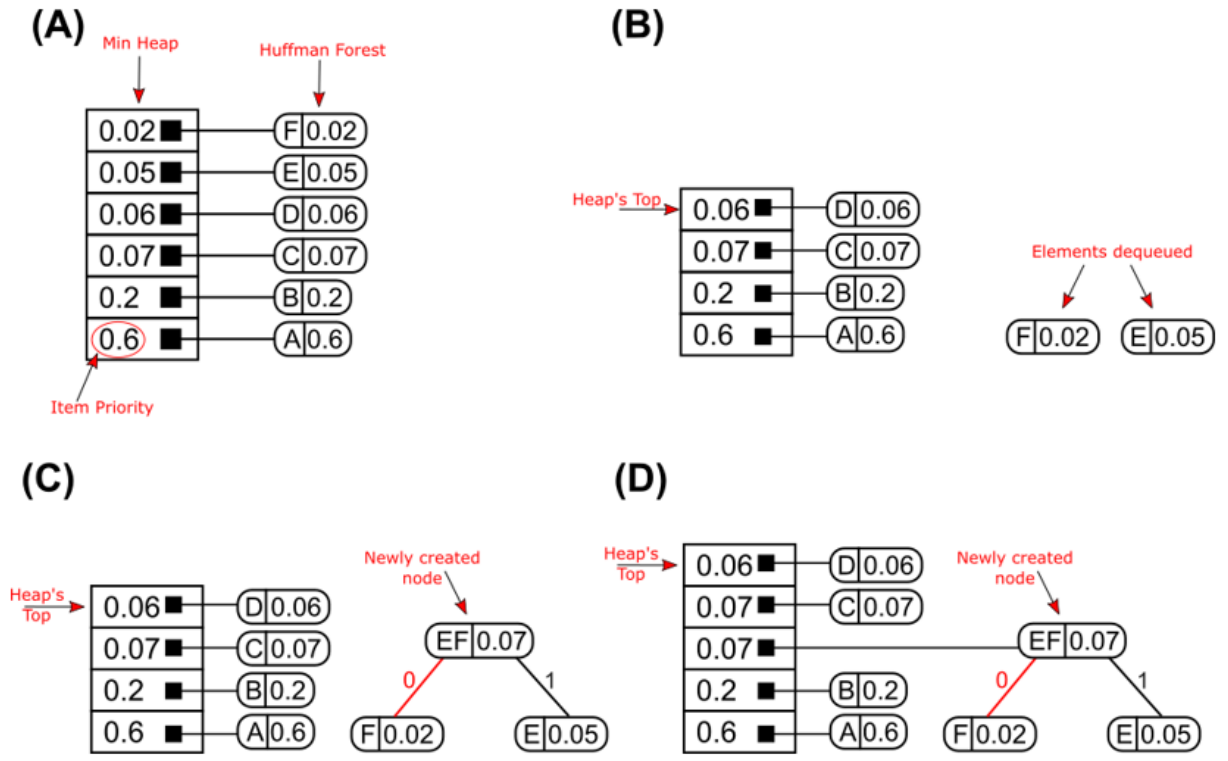


Figure 2.16 The first step in the Huffman coding algorithm. As mentioned in the text, the algorithm will use 2 auxiliary data structures, a priority queue and a binary tree. Each tree node will have a value, a set of characters in the text, and a priority, the sum of the frequencies of those characters in the text.
(A) Initially we create one tree node for each character, associated with its frequency in the text. We also add each node into a priority queue, using the frequency as its priority (smaller frequency means higher priority hence we would use a min-heap).
(B) We extract two elements from the top of the priority queue.
(C) We create a new tree node to which we add the